

PUJITH SAKHAMURI

Full Stack Data & AI Engineer

pujithchowdarys@gmail.com | United States

LinkedIn: [linkedin.com/in/pujith-sakhamuri-06b69a137](https://www.linkedin.com/in/pujith-sakhamuri-06b69a137) | Portfolio: pujith-sakhamuri-portfolio.vercel.app | GitHub: github.com/pujiverse

PROFESSIONAL SUMMARY

Full Stack Data & AI Engineer with expertise building production-grade data platforms, semantic models, and autonomous AI-ready datasets on Google Cloud Platform (GCP). Skilled in orchestrating batch and real-time pipelines using Cloud Dataflow, Pub/Sub, and Cloud Composer, while deploying containerized API serving layers on Cloud Run. Experienced in preparing datasets for Agent Platform(Vertex AI), automating infrastructure with Terraform, implementing GitHub Actions CI/CD workflows, and enforcing rigorous data governance across healthcare and retail enterprise systems.

TECHNICAL SKILLS

- **GCP Cloud Data & Serving:** Google Cloud Storage (GCS), BigQuery, Cloud Dataflow, Cloud Composer (Airflow), Cloud Functions, Cloud Run, Pub/Sub, Cloud IAM, Cloud scheduler, Workflows, Data proc, Data flow, Service accounts, Terraform
- **AI & Semantic Layers:** Agent Platform(Vertex AI) [Model Serving & Data Foundations], Semantic Layer Modeling, Dimensional Modeling (Star/Snowflake), Agentic AI Data Prep
- **Programming & Integration:** Python (Pandas, NumPy, PySpark, FastAPI), SQL (BigQuery SQL, PostgreSQL, MySQL), Shell Scripting (Bash), REST APIs
- **DevOps & Data Governance:** Terraform (IaC), GitHub Actions (CI/CD), Git, Docker, Linux, Data Dictionaries, Metadata Management, Data Quality Frameworks

PROFESSIONAL EXPERIENCE

Network Data Analyst | Telka | Client: T-Mobile

Aug 2026 – Present

- **Drive Test & KPI Analytics:** Processed and visualized large-scale 5G/LTE drive test datasets and RF metrics (RSRP, RSRQ, SINR, throughput) to identify coverage gaps and optimize cell signal quality.
- **Telemetry Data Automation:** Developed Python and SQL scripts to automate ingestion, cleansing, and validation of raw field testing logs into analytical databases.
- **FCC Compliance & Reporting:** Conducted FCC test compliance audits and built weekly KPI tracking dashboards to monitor network cluster readiness and cell handovers.

Full Stack Data & AI Engineer | SR Systems | Client: CVS Pharmacy

July 2024 – July 2026

- **AI-Ready Semantic Modeling:** Translate complex pharmacy claims, prescription, and inventory data into dimensional entities and semantic models in BigQuery to ground Vertex AI workflows and downstream autonomous agent tool calling.
- **Stream & Batch Orchestration:** Engineer robust data pipelines leveraging Cloud Dataflow for streaming/batch transformations, Pub/Sub for real-time messaging, and Cloud Composer (Airflow) for automated DAG scheduling.
- **Serving & API Integration:** Develop and deploy containerized microservices and API serving layers on Cloud Run to expose curated data products to web portals and 3rd-party applications.
- **CI/CD & Infrastructure as Code:** Automate cloud infrastructure provisioning using Terraform and maintain multi-environment CI/CD deployment pipelines (Dev, Stage, Prod) using GitHub Actions.
- **Data Governance & Quality:** Establish enterprise data dictionaries, metadata catalogs, and automated data quality validation frameworks, ensuring full observability and strict HIPAA compliance.

Data Engineer | IvyNova | Client: Cleveland Clinic

June 2023 – July 2024

- **Data Lifecycle & Landing:** Architected automated ingestion pipelines for structured and semi-structured clinical feeds into GCS landing zones, performing discovery and preparation for data science MVPs.
- **Serverless Pipeline Engineering:** Built event-driven serverless Cloud Functions and scheduled Airflow DAGs to automate data transformations, achieving 99.8% data accuracy across clinical staging tables.
- **Dimensional Modeling & Presentation:** Designed normalized and star-schema models in BigQuery, collaborating with clinical teams to map customer data journeys into presentation-ready datasets for BI dashboards.

Data Analytics Specialist (Graduate Assistant) | *Cleveland State University*
Mar 2022 – May 2023

- **Institutional Databases & Reporting:** Managed relational SQL databases tracking enrollment and academic metrics; designed operational Power BI and Tableau dashboards for administrative departments.
- **Workflow Automation:** Automated recurring administrative data collection and reporting workflows using Python, conducting data quality and integrity audits on survey datasets.

Freelance Developer / Data Analyst | *Self-Employed, Hyderabad, India*
Oct 2020 – Jan 2022

- **Student Project Mentoring:** Mentored engineering students on academic projects, from topic selection and solution design to coding, testing and final presentation.
- **Web Application Development:** Designed and built web applications for student and personal projects, covering front-end pages, back-end logic and databases.
- **Debugging & Documentation:** Helped students debug code, write project documentation and prepare for project reviews.

EDUCATION

Master of Science in Artificial Intelligence and Machine Learning | Indiana Wesleyan University (IWU) (July 2026 – Present)
Master of Science in Computer and Information Sciences | Cleveland State University, Cleveland, OH (Jan 2022 – May 2023)
Bachelor of Technology in Computer Science and Engineering | JNTUH, Hyderabad, India (Aug 2016 – May 2020)

FREELANCE & CONTRACT EXPERIENCE

Full Stack & Generative AI Developer | *Upwork & Freelancer.com*
Jan 2024 – Present

- **Full-Stack Application Engineering:** Architected and deployed scalable web applications using Next.js, React, TypeScript, and Python (FastAPI/Flask), establishing CI/CD automation targeting Vercel and Google Cloud Run.
- **Generative AI Tooling Integration:** Integrated Gemini API, open-source LLMs, and browser APIs to engineer AI tools featuring structured JSON output parsing, automated speech synthesis, and media processing.
- **Operational Solutions & Dashboards:** Delivered tailored digital systems for small-to-medium businesses, constructing dynamic admin dashboards, customized UI components, and integrated messaging via WhatsApp Business API.
- **Architecture & Cloud Strategy Consulting:** Provided end-to-end technical consulting for non-technical founders, optimizing tech stack selection, database schema design, and serverless infrastructure to lower recurring operational overhead.

AI & LLM Evaluation Specialist | *Alignerr*
Jan 2025 – Present

- **Model Evaluation & RLHF:** Benchmarked and evaluated LLM outputs across complex Python scripts, SQL data transformations, and multi-step reasoning tasks under rigorous RLHF guidelines.
- **Agentic Data Engineering:** Authored and validated multi-turn system prompts and tool-calling validation datasets to drive deterministic execution across autonomous agent frameworks and RAG pipelines.
- **Ground-Truth Validation & Quality Assurance:** Curated structured ground-truth response pairs and audited multimodal AI task completions, actively mitigating model hallucinations and improving prompt fidelity.

FEATURED PROJECTS & DEPLOYMENTS

BizManager Pro – SaaS Business Operations Platform | *Next.js 15, React 19, TypeScript, Tailwind CSS, Python FastAPI, PostgreSQL, Supabase, Vercel*
<https://biz-manager-eight.vercel.app/>

- **Architecture & Development:** Engineered a full-stack SaaS platform managing business inventory, invoicing, CRM, and operational analytics with real-time sync.
- **Serving & Analytics:** Integrated automated reporting dashboards and REST API services for seamless third-party client management.

Sai Indian Restaurant – Digital Ordering & Admin System | *React, TypeScript, Tailwind CSS, Python FastAPI, WhatsApp API, Google Cloud Run*
<https://sai-indian.vercel.app/>

- **Full-Stack Implementation:** Built a complete web portal featuring an interactive menu, automated table reservations, and online order processing.

- **API Messaging Integration:** Connected WhatsApp Business API for instant automated customer notifications and order confirmations.

Smart Scene Cutter – AI Video Segmentation Tool | *Python, PyTorch, OpenCV, FFmpeg, Streamlit, Docker, Cloud Run*
<https://smart-scene-cutter.vercel.app/>

- **AI Video Processing:** Developed computer vision algorithms for automated scene boundary detection and timestamp-based video splitting.
- **Cloud Deployment:** Containerized processing pipelines into Docker images and deployed on Google Cloud Run for auto-scaling execution.

Multimodal Generative AI Studio Suite | *React, TypeScript, Gemini API, Python FastAPI, Web Audio API, Canvas API*
<https://ai-multimodal-studio.vercel.app>

- **Generative AI Integration:** Built an end-to-end studio combining text, audio, and visual AI models powered by Gemini API and custom browser pipelines.
- **Structured Processing:** Implemented structured JSON prompt orchestration to drive real-time multi-modal content generation.

AI Animated Presentation Generator | *Next.js, TypeScript, Gemini 2.5 API, Remotion, Tailwind CSS, Vercel*
<https://pujiverse-ai-animated-presentation.vercel.app/>

- **Automated Slide Rendering:** Architected an automated generator creating styled presentation decks and animations via programmatic video rendering.
- **LLM Orchestration:** Leveraged Gemini APIs to convert raw user prompts into structured slide layouts and narrative flow.

AI Resume Builder & ATS Job Matcher | *React, TypeScript, Python, Gemini API, PDFKit, Tailwind CSS*
<https://resume-builder-azure-omega.vercel.app/>

- **ATS Resume Optimization:** Engineered an AI platform parsing job descriptions against resumes to calculate match scores and highlight skill gaps.
 - **Document Generation:** Integrated automated PDF generation tools with optimized formatting for applicant tracking systems.
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